

24/4/26

29/4/26

LECTURE - 22 & 23

Wednesday
Friday.

One & Two sided Confidence Interval: For
Testing of Hypothesis for Population Mean.

① Two sided:

1) Formulation of hypothesis:

$$H_0: \mu = \mu_0$$

(μ₀ is the one given in ques.)

$$H_1: \mu \neq \mu_0$$

2) $\alpha = 0.05$ (if not given)

3) Test-statistic: Z-test OR T-test

4) Calculate C.I: No calculate

For Z-test: $\alpha \rightarrow \alpha/2 \rightarrow 1 - \alpha/2 \rightarrow$ answer } Find corresponding Z, ^{level of significance} table ⁱⁿ _(μ)
For t-test: $\alpha \rightarrow \alpha/2 \rightarrow$ answer } t value to answer

5) Conclusion: Reject H₀ if μ₀ doesn't falls in range of C.I

② Left-Tailed:

$$H_0: \mu = \mu_0$$

$$H_1: \mu < \mu_0$$

2) ~~α = 0.05~~ α = 0.05 (if not given)4) For Z-test: $\alpha \rightarrow 1 - \alpha =$ answerFor t-test: $\alpha =$ answer5) Reject H₀, if μ₀ < C.I (μ)

$$3) Z\text{-test} = \bar{X} + Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$$

$$t\text{-test} = \bar{X} + t_{\alpha(n)} \cdot \frac{s}{\sqrt{n}}$$

③ Right-Tailed:

$$1) H_0: \mu = \mu_0$$

$$H_1: \mu > \mu_0$$

2), ~~3~~ 4) Same5) Reject H₀, if μ₀ > C.I (μ)

$$3) Z\text{-test} = \bar{X} - Z_{\alpha} \cdot \frac{\sigma}{\sqrt{n}}$$

$$t\text{-test} = \bar{X} - t_{\alpha(n)} \cdot \frac{s}{\sqrt{n}}$$

Q: Internet Service Provider claims its service delivers same speed of 20Mbps. To verify this claim, a random sample of 32 users was taken which showed mean speed of 18Mbps with a SD of 2Mbps at 5% level of significance. Test whether actual mean speed is ^{left-tailed} less than the claimed 20Mbps.

① $H_0: \mu = 20\text{Mbps}$

$H_1: \mu < 20\text{Mbps}$

② $\alpha = 0.05$

③ Z-test (as ~~unknown~~ but $n > 30$)

④ $\alpha = 0.05 \rightarrow 1 - \alpha = 0.95$

$Z_{0.95} = 1.645$

$\bar{X} \pm Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}} = \text{Left-tailed C.I.}$

$18 + (1.645) \cdot \frac{2}{\sqrt{32}} = \boxed{18.58}$

⑤ ~~Since~~ Since $\mu_0 = 20 \neq 18.58 \therefore$ we won't reject H_0 (i.e: $\mu = 20$) means supplier made the right claim.

Q: $\mu_0 = 2$, $n = 20$, ~~sample~~ $\bar{X} = 1.5$, $\sigma = 1$.

(a) Test claim of $\mu_0 = 2$ at significance level of 0.01.

* Two tailed: $H_0: \mu = 2$, $\alpha = 0.01$, Z-test (as σ known).
 $H_1: \mu \neq 2$

$\alpha = 0.01$
 $\alpha/2 = 0.005$
 $1 - \alpha/2 = 0.995$

$Z_{0.995} = \boxed{2.57}$

$\mu = \bar{X} \pm Z_{\alpha/2} \cdot (\sigma / \sqrt{n})$
 $\mu = 1.5 \pm (2.57) (1 / \sqrt{20})$
 $\boxed{0.925 < \mu < 2.074}$

answer: Since μ_0 (i.e: 2) is in the range of C.I (μ) $\therefore$ ACCEPTED.

(b) Test claim if SD is unknown & Sample SD is 1 $\rightarrow S$

$\alpha = 0.01$

$\alpha/2 = 0.005$ (as two tailed)

$t_{0.005} = \boxed{2.861}$ $\xrightarrow{\text{after T-test}}$ $0.941 < \mu < 2.059$

answer: since $\mu_0 = 2$ is in range of $\mu \therefore$ ACCEPTED.